

Lecture - 13

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

TEST OF HYPOTHESIS FOR POPULATION CORRELATION COEFFICIENT

We have already discussed the concept of correlation. Where, we studied that if two variables are related in such a way that change in the value of one variable affects the value of another variable then the variables are said to be correlated or there is a correlation between these two variables. Correlation can be positive, which means the variables move together in the same direction, or negative, which means they move in opposite directions. And correlation coefficient is used to measure the intensity or degree of linear relationship between two variables. The value of correlation coefficient varies between -1 and $+1$, where -1 representing a perfect negative correlation, 0 Small Sample Tests representing no correlation, and $+1$ representing a perfect positive correlation.

Sometime, the sample data indicate for non-zero correlation but in population they are uncorrelated ($\rho = 0$).

For example, price of tomato in Delhi (X) and in London (Y) are not correlated in population ($\rho = 0$). But paired sample data of 20 days of prices of tomato at both places may show correlation coefficient (r) $\neq 0$. In general, in sample data $r \neq 0$ does not ensure in population $\rho \neq 0$ holds.

In this section, we will know how we test the hypothesis that population correlation coefficient is zero.

Assumptions

This test works under following assumptions:

- (i) The characteristic under study follows normal distribution in both the populations. In other words, both populations from which random samples are drawn should be normal with respect to the characteristic of interest.
- (ii) Samples observations are random.

Let us consider a random sample $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ of size n taken from a bivariate normal population. Let ρ and r be the correlation coefficients of population and sample data respectively.

Here, we wish to test the hypothesis about population correlation coefficient (ρ), that is, linear correlation between two variables X and Y in the population, so we can take the null hypothesis as

$$H_0: \rho = 0 \text{ and } H_1: \rho \neq 0 \text{ [for two – tailed test]}$$

[Here $\theta = \rho$ and $\theta_0 = 0$ if we compare it with general procedure]

Or

$$\left. \begin{array}{l} H_0: \rho \leq 0 \text{ and } H_1: \rho > 0 \\ H_0: \rho \geq 0 \text{ and } H_1: \rho < 0 \end{array} \right\} \text{ [for one - tailed test]}$$

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{(n-2)}$$

which follows t-distribution with $n - 2$ degrees of freedom.

After substituting values of r and n , we find out calculated value of test statistic t . Then we look for critical (or cut-off or tabulated) value(s) of test statistic t from the t -table. On comparing calculated value and critical value(s), we take the decision about the null hypothesis as discussed.

Let us do some examples of testing of hypothesis that population correlation coefficient is zero.

Example 1: A random sample of 18 pairs of observations from a normal population gave a correlation coefficient of 0.7. Test whether the population correlation coefficient is zero at 5% level of significance.

Solution: Given that

$$n = 18, r = 0.7$$

Here, we wish to test that population correlation coefficient (ρ) is zero so our claim is $\rho = 0$ and its complement is $\rho \neq 0$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0: \rho = 0 \text{ and } H_1: \rho \neq 0$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, we want to test the hypothesis regarding population correlation coefficient is zero and the populations under study follow normal distributions, so we can go for t-test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.7\sqrt{18-2}}{\sqrt{1-(0.7)^2}} = \frac{0.7 * 4}{\sqrt{0.71}} = 3.94$$

The critical value of test statistic t for two-tailed test corresponding $(n-2) = 16$ df at 5% level of significance are $\pm t_{(n-2), \alpha/2} = \pm t_{(16), 0.025} = \pm 2.120$

Since calculated value of test statistic t ($= 3.94$) is greater than the critical values ($= \pm 2.120$), that means calculated value of test statistic t lies in rejection region, so we reject the null hypothesis. i.e. we reject the claim at 5% level of significance.

Thus, we conclude that sample provides us sufficient evidence against the claim so there exists a relationship between two variables.

In the same way, you can try the following exercise.

- (i) Twenty families were selected randomly from a colony to determine that correlation exists between family income and the amount of money spent per family member on food each month. The sample correlation coefficient was computed as $r = 0.40$. Assuming that the family income and the amount of money spent per family member on food each month follow normal distributions then test that there is a positive linear relationship between the family income and the amounts of money spent per family member on food each month in colony at 1% level of significance.
- (ii) A random sample of 15 married couples was taken from a population consisting of married couples between the ages of 30 and 40. The correlation coefficient between the IQs of husbands and wives was found to be 0.68. Assuming that the IQs of husbands and wives follow normal distributions then test that IQs of husbands and wives in the population are positively correlated at 1% level of significance.